Petrochemical Technology

Feedstocks for Petrochemicals

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Gas:

Natural gas, Associated Gas, Gas condensate, Refinery off gas

Light Liquids:

Natural gas liquids, Naphtha, Reformate, Kerosene

Heavy Liquids:

Residuum, Low sulphur heavy stocks, Fuel oil, etc.

All feedstocks require certain type of purification.

Maximum purification is required for gaseous feedstocks as they are obtained either from a field or from a process.

Natural gas production in 2021-22: 34.02 BCM
NG Consumption: 63.91 BCM

Typical Composition of Bombay High Gas:
CH_4: 78.48; C_2H_6: 7.24; C_3H_8: 4.59; C_4H_{10}(i- & n-): 1.95; C_5^+: 0.5;
CO_2: 6.45; H_2S+RSH: 0.78; N_2+O_2: 0.01

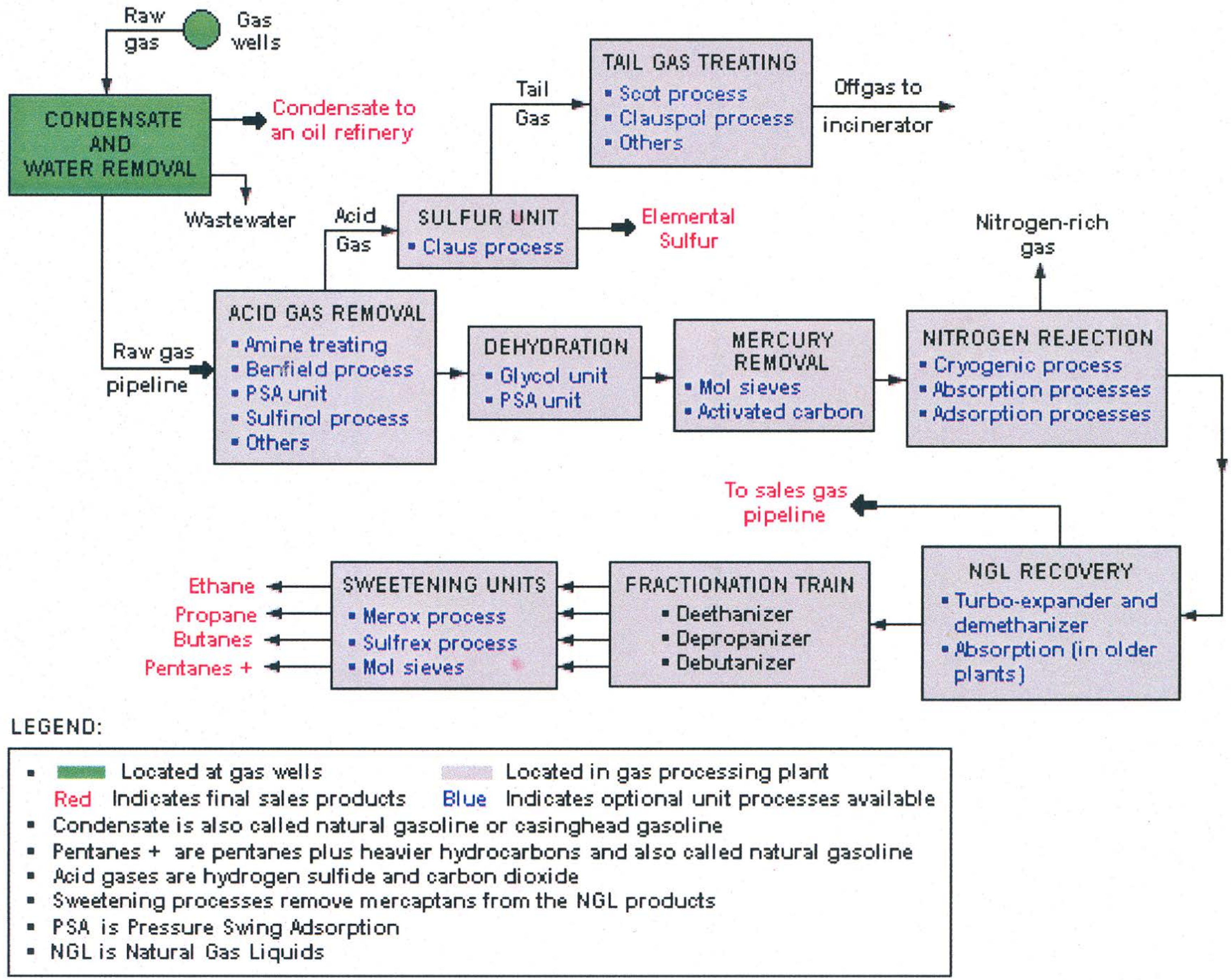
Most Common Impurities of Gaseous Feedstock

Water vapour: Forms crystalline hydrates ($CH_4.7H_2O$, $C_2H_6.7H_2O$, etc.)
- stable solids below 19 ^oC

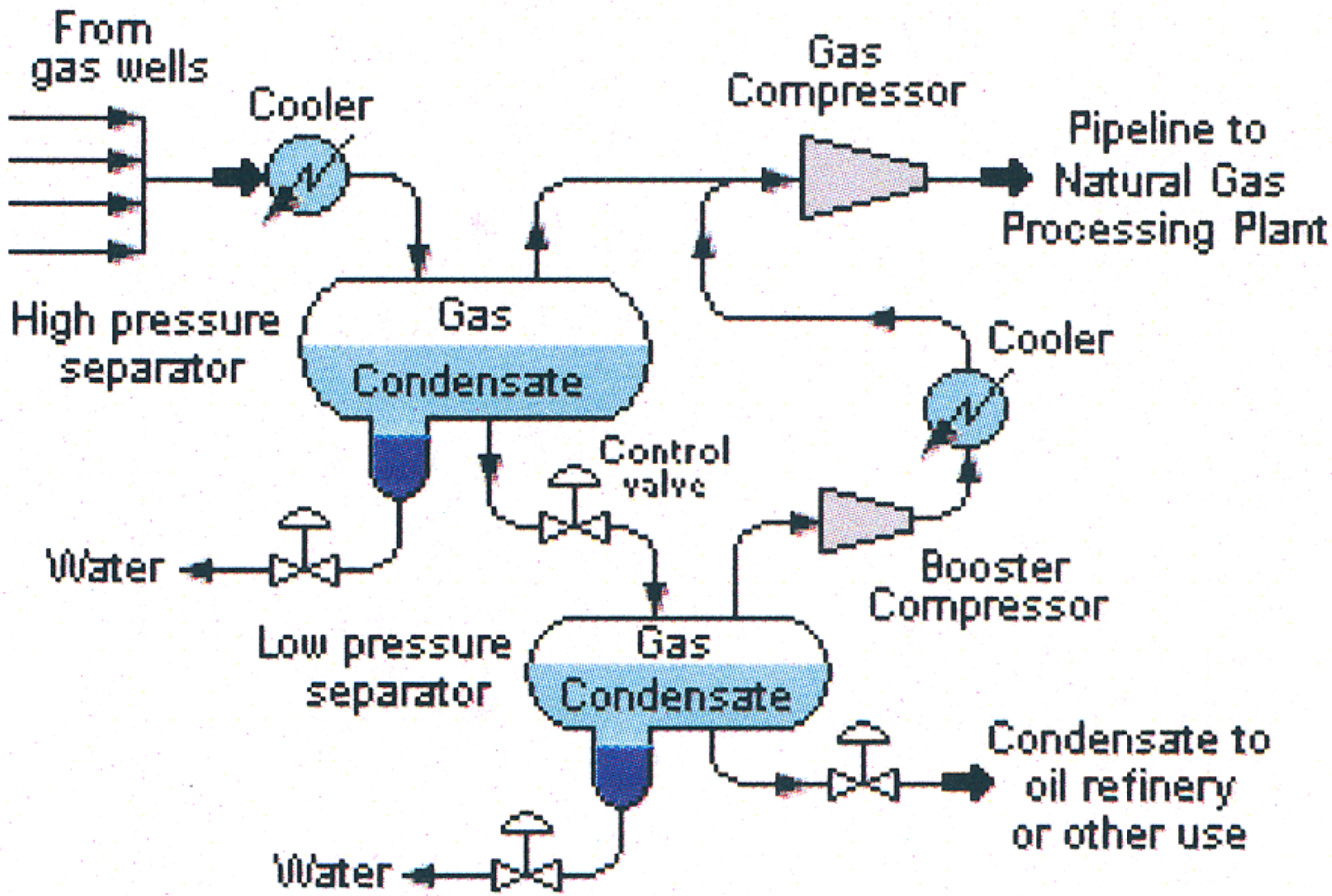
Removed by passing through drying agents

Mechanical (suspended) impurities: Clay, dust, catalyst particles, tar droplets, etc.
Mostly removed by washing with water
Suitable solvent required for removal of tar and heavy oils

Chemical impurities: CO_2, COS, H_2S, RSH, NH_3, etc.
Mostly removed by absorption

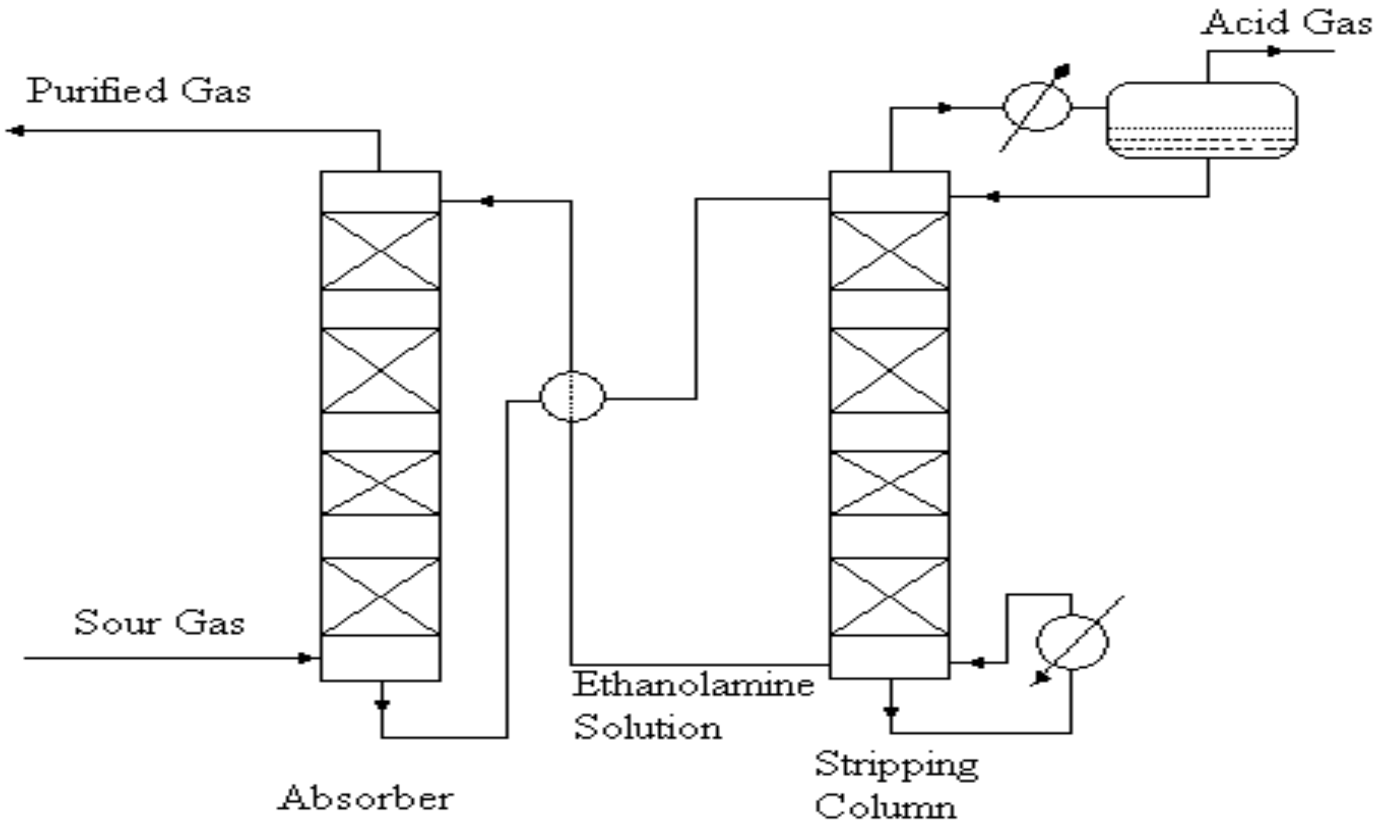


Schematic Flow Diagram for the Processing of Natural Gas



Schematic Flow Diagram of the Separation of Condensate from Raw Natural Gas

Amine Treating Unit



Claus Process

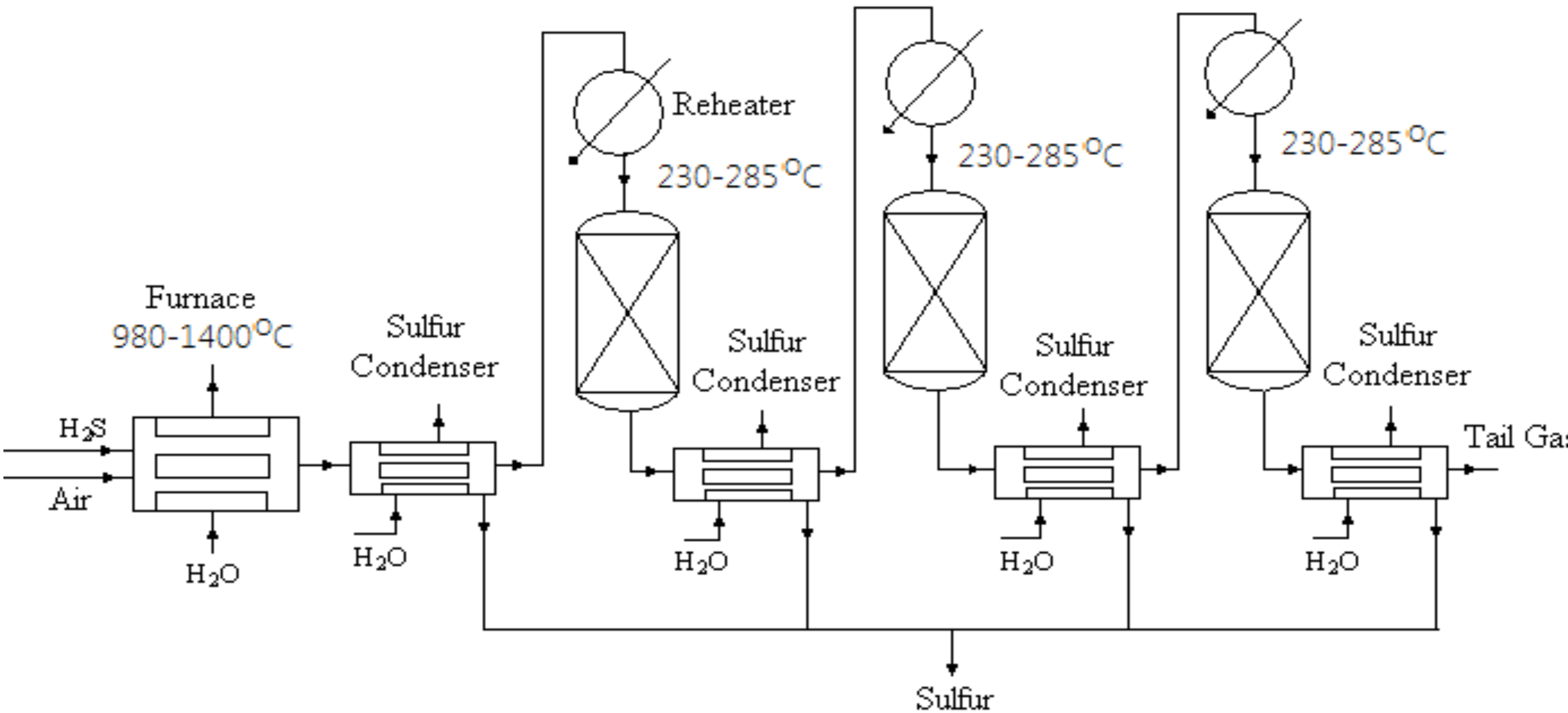
Thermal Stage

$H_2S + 3/2O_2 = H_2O + SO_2 + \Delta$

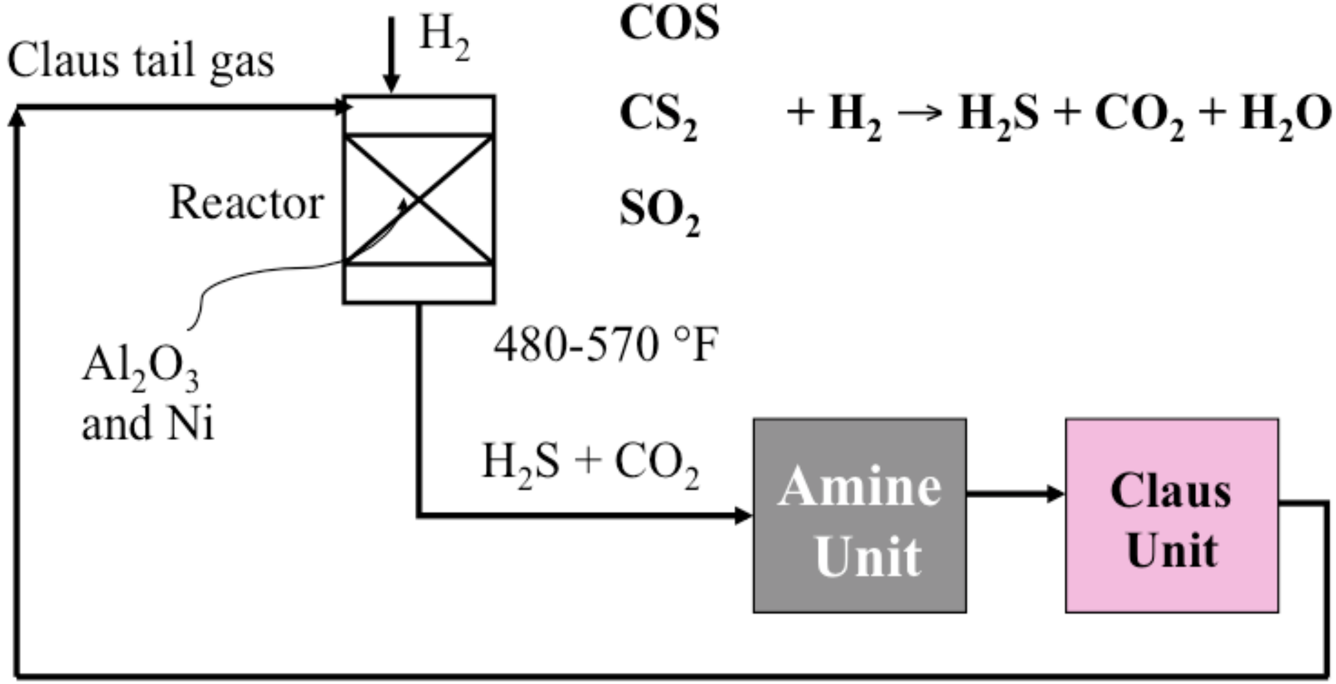
$2H_2S + SO_2 = 2H_2O + 3S + \Delta$

Catalytic Stage

$2H_2S + SO_2 = 2H_2O + 3S + \Delta$



Sulfur Recovery -SCOT Process

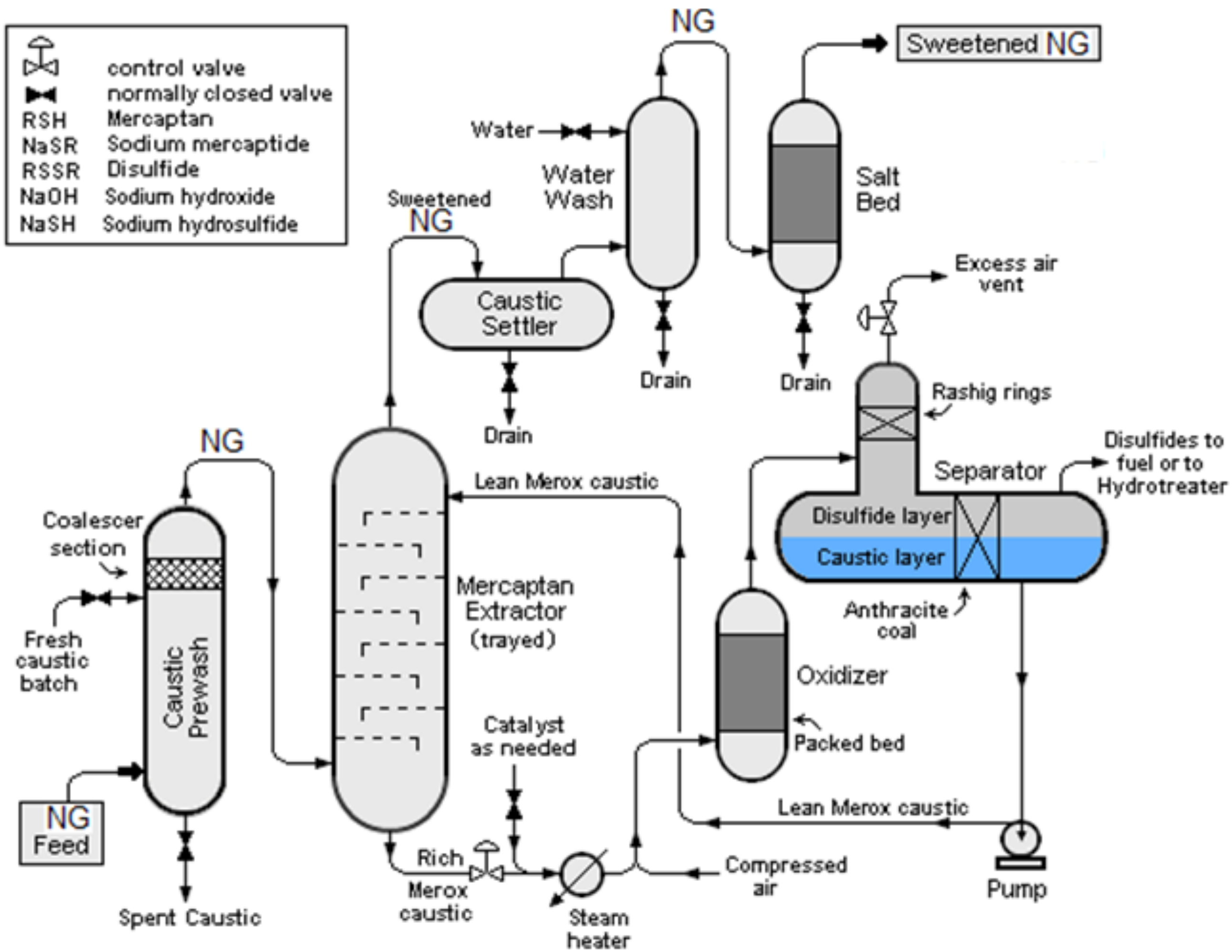


$$\left.\begin{array}{l}\mathbf{COS}\\ \mathbf{CS_2}\\ \mathbf{SO_2}\end{array}\right. + \mathbf{H_2} \rightarrow \mathbf{H_2S + CO_2 + H_2O}$$

Claus +SCOT Recover greater than 99% of S.

The Shell Claus Off-gas Treatment (SCOT) Process

MerOx Process



CHEMICAL REACTIONS IN MEROX TREATING

Caustic Prewash:
$NaOH + H_2S \rightarrow NaSH + H_2O$

Extraction:
$2RSH + 2NaOH \rightarrow 2NaSR + 2H_2O$

Overall Mercaptan Conversion:
$4RSH + O_2 \rightarrow 2RSSR + 2H_2O$

Regeneration:
$4NaSR + O_2 + 2H_2O \rightarrow 2RSSR + 4NaOH$

Catalyst: Cobalt phthalocyanine monosulphate

Liquefied Natural Gas (LNG)

[-163^{o}C close to atmospheric pressure (3.6 psig)]

LNG takes 1/600th volume of NG

LNG is transported in cryogenic vessels or cryogenic road tankers

Compressed Natural Gas (CNG)

NG at 20-25MPa pressure at ambient temperature
(Compressed to <1% of its volume at STP)

Processes for LNG Production

(1) APCI Propane Pre-cooled Mixed Refrigerant Process

(2) Philips Optimized Cascade process

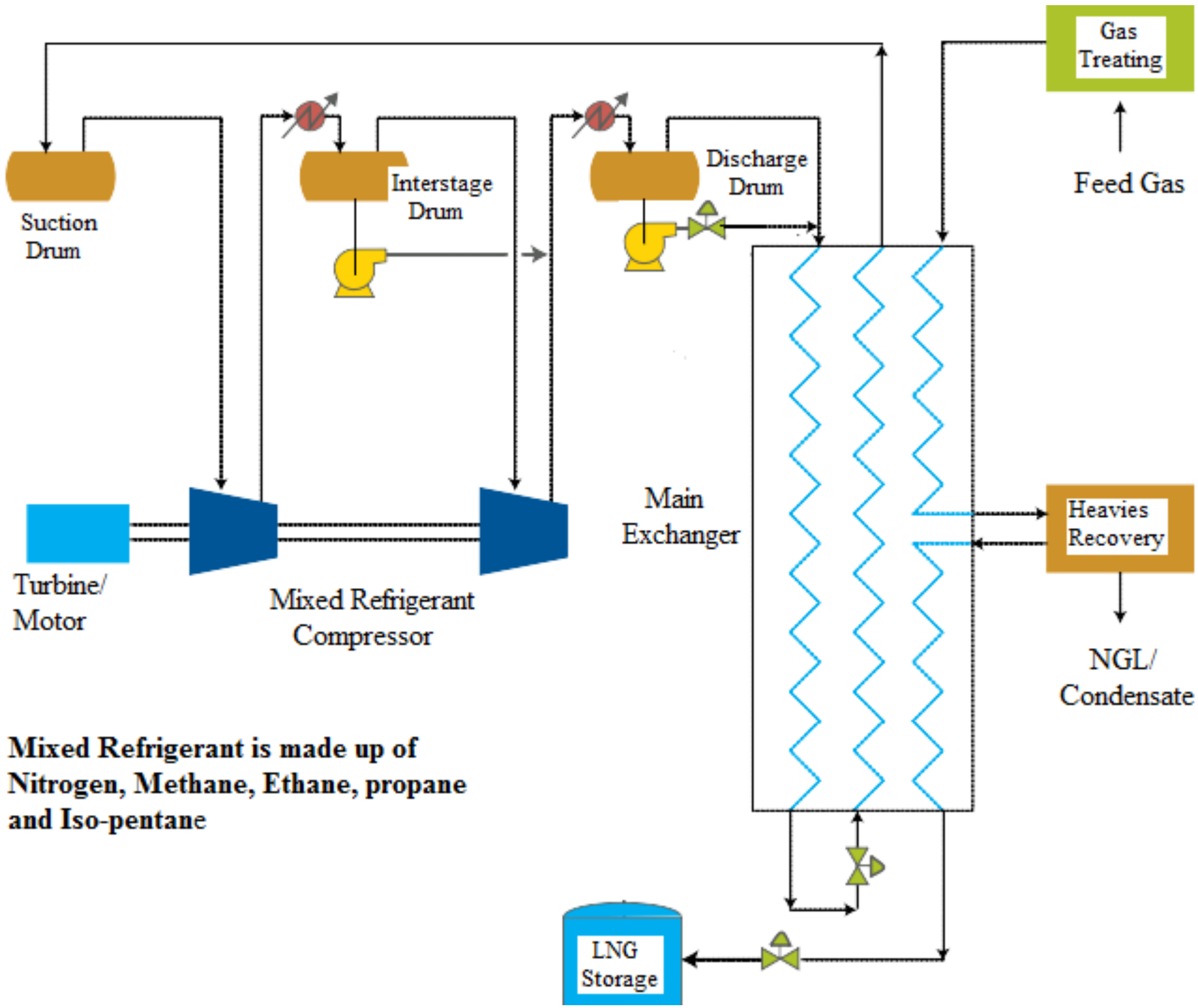
(3) Black & Veatch PRICO Process

(4) Statoil/Linde Mixed Fluid Cascade Process

(5) Axens Liquefin Process

(6) Shell Double Mixed Refrigerant Process

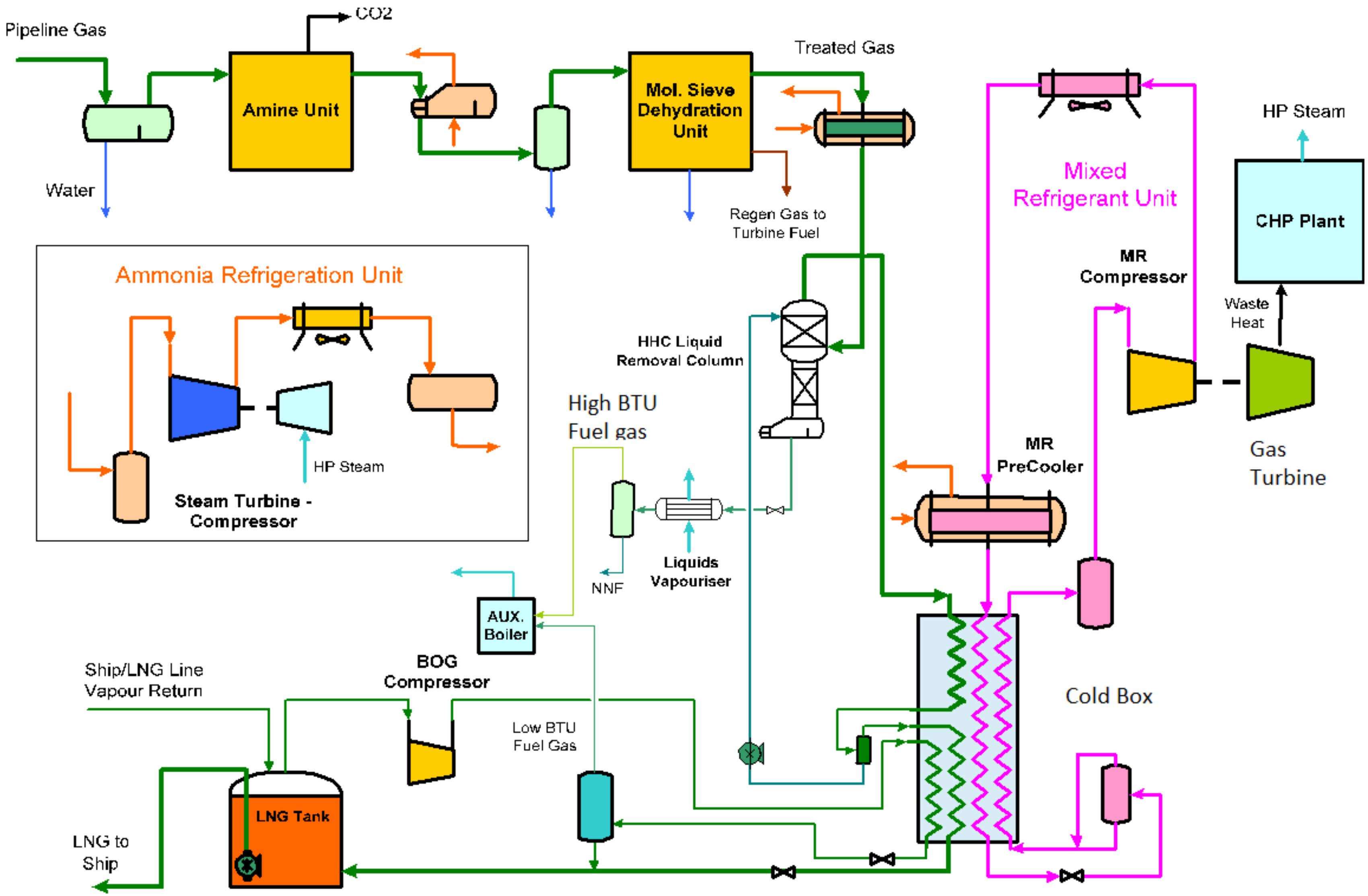
(7) OSMR (Optimized Single Mixed Refrigerant) Process



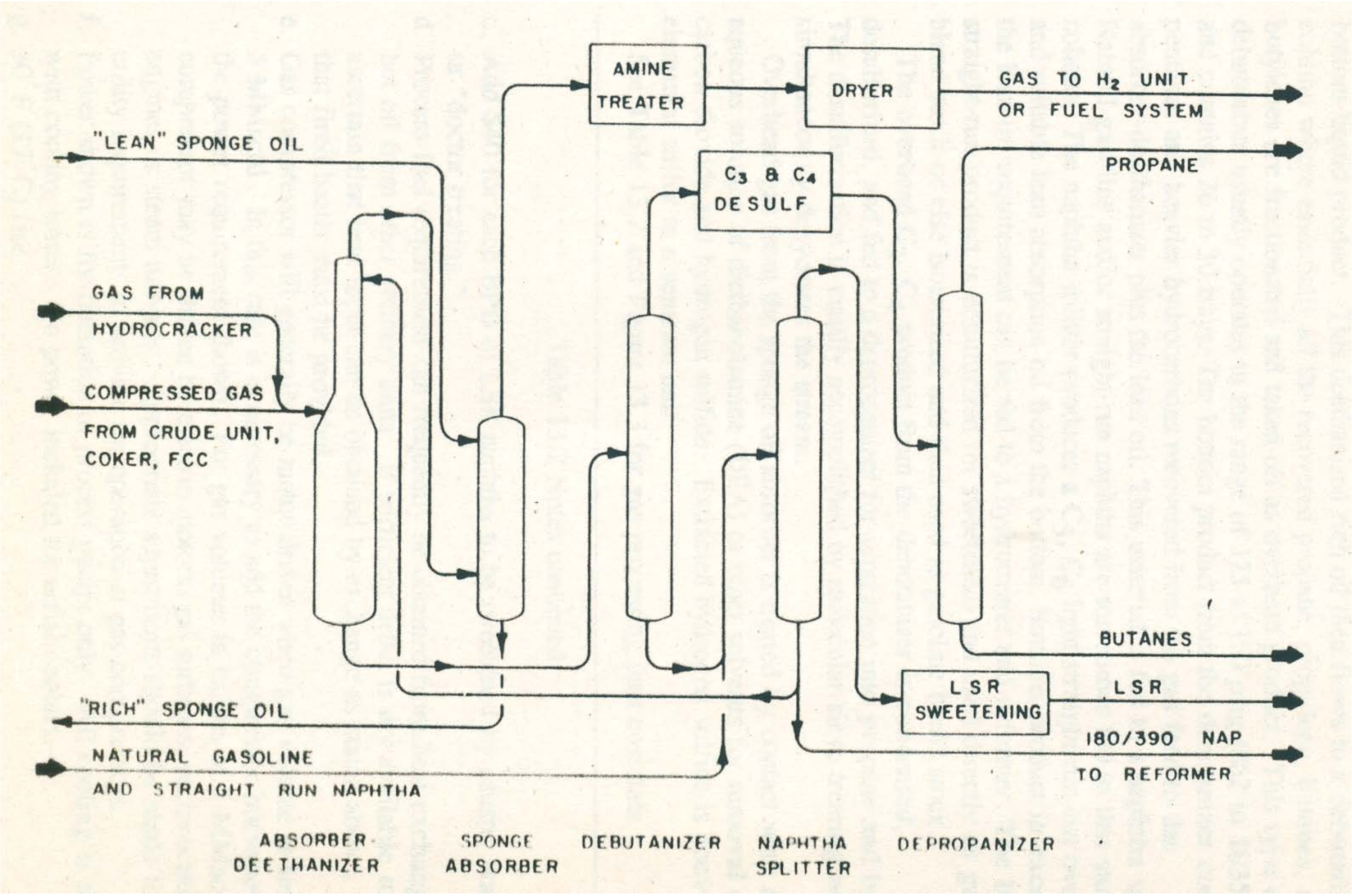
Mixed Refrigerant is made up of Nitrogen, Methane, Ethane, propane and Iso-pentane

Poly Refrigerant Integrated Cycle Operation (PRICO®) Process

By **Black & Veatch Corpn.**



Optimized Single Mixed Refrigerant (OSMR) Process



Flow Diagram of Refinery Off Gas Processing

Treatment of Liquid Feedstocks

Hydrotreating of Naphtha

Heavy naphtha is catalytically reformed to produce aromatics.

The active material in most catalytic reforming catalyst is platinum. Certain metals, hydrogen sulfide, ammonia, and organic nitrogen and sulfur compounds will deactivate the catalyst. Feed processing, in the form of hydrotreating, is usually employed to remove these materials.

1. Desulphurization:

Mercaptans: $RSH + H_2 \rightarrow RH + H_2S$

Sulphides: $R_2S + 2\ H_2 \rightarrow 2RH + H_2S$

Disulphides: $(RS)_2 + 3\ H_2 \rightarrow 2RH + 2\ H_2S$

Thiophenes: $C_4H_4S + 4\ H_2 \rightarrow C_4H_{10} + H_2S$

2. Denitrogenation:

Pyrrole: $C_4H_4NH + 4\ H_2 \rightarrow C_4H_{10} + NH_3$

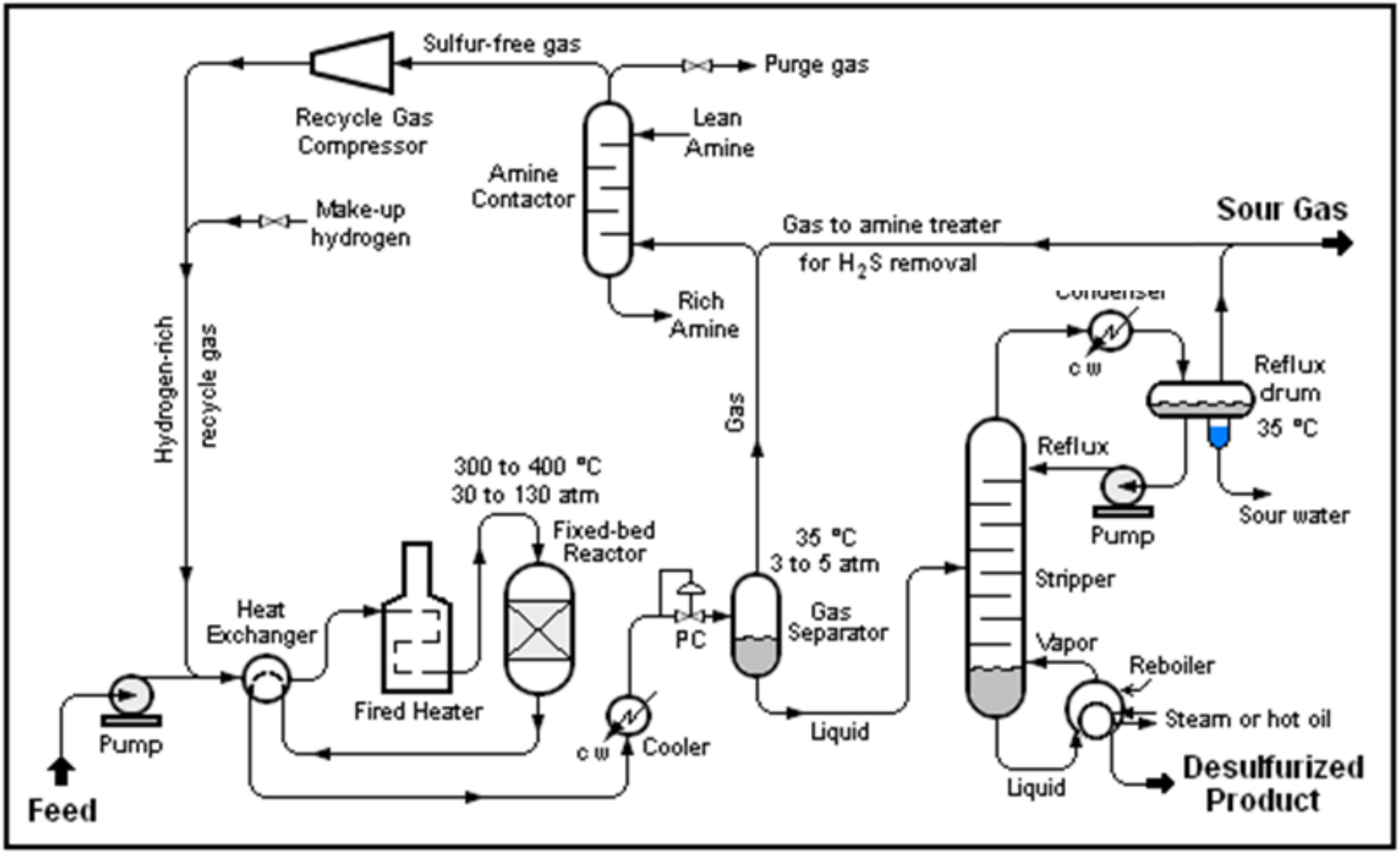
Pyridine: $C_5H_5N + 5\ H_2 \rightarrow C_5H_{12} + NH_3$

Process Conditions:

Temperature: 300 – 400 ^{o}C

Pressure : 1000 – 3000 psig

Catalyst: Co-Mo/Al_2O_3, Ni-Mo/ Al_2O_3, W-Mo/ Al_2O_3, Ni-Co-Mo/ Al_2O_3



Hydrotreating Process Schematic

Thank you